



Impact of photobiomodulation for oral mucositis on body weight and BMI of patients with head and neck cancer

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Abstract

Purpose To evaluate the impact of photobiomodulation for oral mucositis on body weight and body mass index (BMI) in head and neck cancer patients.

Methods Patients were separated into two groups: photobiomodulation (PBM) ($n = 17$) and SHAM ($n = 19$). Participants assigned to the PBM group received the photobiomodulation therapy on alternate days at 33 different sites of the oral mucosa (1 J for 10 s per application point) using a low-power aluminum gallium arsenide (AsGaAl) laser device with a maximum output power of 86.7 mW and a wavelength of 660 nm (red). The same protocol was performed in patients allocated in the SHAM group but with the device turned off. At the first and twenty-fourth radiotherapy sessions, participants from both groups underwent an intraoral examination to assess the presence or absence of mucositis using the WHO scale, and measurements of weight and height for BMI calculation were undertaken.

Results All patients had significant differences in weight and BMI throughout the study period. Patients in the PBM group lost less weight ($p < 0.01$) and had a reduced BMI loss ($p < 0.01$) in comparison to patients in the SHAM group. Although PBM did not prevent oral mucositis, it decreased its severity ($p < 0.01$).

Conclusion Photobiomodulation for oral mucositis reduced weight loss and prevented a reduction in BMI in patients who underwent chemoradiotherapy for head and neck cancer. Retrospectively registered at <https://data.mendeley.com/datasets/4kd7s49wk4/1>.

Keywords Body weight · BMI · Photobiomodulation · Head and neck cancer · Nutritional status

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Introduction

Accelerated weight loss and malnutrition are secondary disorders that frequently affect cancer patients, and their occurrence can vary on average from 40 to 80% in individuals diagnosed with head and neck cancer (HNC) [1]. Both secondary disorders are caused by local and metabolic alterations of a vast magnitude produced by the tumor cells and their growth and the side effects of antineoplastic treatments [2–6].

The primary treatment modalities for HNC are surgery, radiotherapy, and chemotherapy, associated with acute side effects, including oral mucositis (OM). OM initially presents as erythema that may progress to erosion and ulceration; it is associated with pain, dysphagia, and odynophagia and can cause speech problems [7]. This set of manifestations can lead to dehydration due to lower food intake and difficulty in fluid intake, resulting in significant weight loss, negative variation in the BMI, and worsening the individual's nutritional status [8].

Prevention and management of OM are required for the patient's optimal quality of life and nutritional status. Laser therapy or PBM effectively relieves pain and decreases the inflammatory processes, which are some of the OM characteristics, as it can regulate various metabolic events through photophysical, photobiological, and biochemical processes [9–11]. Thus, PBM could reduce weight loss and BMI lowering by preventing or treating OM.

Anthropometric measurements can be used to assess the impact of OM on the nutritional status of individuals [1]. Weighing and measuring a patient's height and body composition at the beginning, during, and at the end of treatment are simple and cost-effective ways to monitor weight loss and nutritional status [12, 13].

This study aimed to evaluate the impact of PBM for OM prevention and treatment on weight loss and BMI in patients diagnosed with HNC at the beginning and the end of antineoplastic treatment.

Methods

This was a prospective randomized clinical study that recruited 36 patients diagnosed with malignant neoplasms in the head and neck region that were to undergo radiotherapy alone or concomitant with other therapies at a High Complexity Oncology Unit (Unidade de Alta Complexidade em Oncologia-UNACON) in Salvador, Bahia, Brazil, from August 2017 to March 2020. The study's ethics approval was obtained from the National Human Research

Ethics Committee (CAAE: 68,263,317.8.0000.0047). The data supporting the study findings are available in Mendeley Data at <https://data.mendeley.com/datasets/4kd7s49wk4/1>, reference [https://doi.org/10.176332/4kd7s49wk4.1].

Inclusion criteria included adult patients who were 18 years of age or older and diagnosed with malignant neoplasms in the head and neck region but had not initiated the proposed treatment protocol. Researchers excluded patients with diabetes mellitus and autoimmune diseases because these conditions impair tissue repair. Also excluded were patients who failed to complete the full research protocol.

To minimize the likelihood of developing OM, patients who agreed to participate in the study were strictly instructed on how to perform oral hygiene with a toothbrush, a non-abrasive toothpaste, and dental floss to maintain their oral health. They were also instructed to increase fluid intake. Before initiating the oncological therapy, patients who required oral care were subjected to dental procedures such as tooth extractions, scaling and root planing, and endodontic treatment.

A socioeconomic questionnaire was used to collect individual data about the patient's lifestyle, treatment modalities (radiotherapy with chemotherapy/surgery), tumor location, and staging. Anthropometric measurements were mandatory throughout the treatment, using a Welmy® anthropometric scale. The BMI was calculated by measuring weight and height, using the formula $BMI = \text{weight}/\text{height}^2$. The results were categorized following the World Health Organization (WHO 2006) [14] classification for adults and the Pan American Health Organization (PAHO) (2002) [15] classification for elderly individuals (over 60 years of age).

Using the classification criteria of Blackburn et al. 1977 [16], the percentage of the speed of involuntary weight loss (SIWL) over time was calculated with the following formula:

$$SIWL(\%) = \frac{\text{Initial weight (kg)} - \text{current weight (kg)} \times 100}{\text{Initial weight (kg)}}$$

Patients were blinded and randomly allocated into two groups: PBM ($n = 17$) and SHAM ($n = 19$) using their file registration number. After randomization and at the start of the experiment, those assigned even numbers were automatically allocated to the PBM group, and those with odd numbers were allocated to the SHAM group.

Participants assigned to the PBM group received the photobiomodulation therapy on specific sites of the oral mucosa, three times a week on alternate days, using the low-power aluminum gallium arsenide (AsGaAl) laser device (Twin Flex®, MM Optics, São Carlos, Brazil) with a maximum output power of 86.7 mW, active tip area of 0.03 cm² (measurement performed by the research team = 0.1246 cm²), and

wavelength of 660 nm (red). The dosimetry used at each application point was 1 J for 10 s, with a total energy of 33 J. Thirty-three equidistant laser application spots were outlined in the oral cavity following the protocol proposed by Scully (2009) [17]: upper and lower labial mucosa (03 points each), right and left buccal mucosa (03 points each), hard and soft palate (03 points each), the anterior floor of the mouth (01 point), right and left lateral borders of the tongue (03 points on each side), dorsal surface of the tongue (06 points), and oral commissures (02 points). The tumor site was not included in the protocol with a safety margin of at least 5 cm.

The same protocol was performed in patients allocated in the SHAM group but with the device turned off. All patients were blinded. At the first and 24th radiotherapy sessions, patients were weighed and submitted to an intraoral examination to evaluate the presence or absence of OM. The severity of OM was assessed according to the WHO scale (1979) [18]. Patients who had their cancer treatment interrupted for any reason were automatically excluded from the study.

The free software R, version 4.0.1 (2020), was used for the statistical analysis of the data. The Shapiro–Wilk test was applied to verify the normality of data distribution. In the intragroup analysis, the Student *t*-test for paired samples was used in the case of continuous variables, and the McNemar–Bowker test for categorical data. For intergroup comparisons, the Student *t*-test was used for continuous data, the Mann–Whitney test for continuous data with a non-normal or ordinal distribution, and Fisher’s exact test for categorical data. Comparisons were conducted between the start of the oncologic treatment (1st session) and the end (24th session). The significance level used for this study was 5%.

Results

The socioeconomic analysis demonstrated that the mean age of the patients was 54.4 years (± 13.06). Most participants were male, married, and self-declared brown, had primary education, a family monthly income of up to 01 minimum wage, and had a history of smoking and alcohol consumption. The oropharynx was the most common tumor location, diagnosed in stage IV, and the majority of patients were under a therapeutic regimen of radiotherapy concomitant with Cisplatin chemotherapy every 21 days (Table 1).

Throughout the study, it was observed that all patients in both groups lost weight and reduced their BMI. The PBM group had a statistically significant mean weight loss and BMI reduction of 2.72 kg (± 2.2) ($p < 0.001$) and 1.05 kg/m² (± 0.83) ($p < 0.001$), respectively, when the 1st and 24th radiotherapy sessions were evaluated and compared. The SHAM group had a significantly higher mean weight loss

Table 1 Socioeconomic, lifestyle, tumor characteristics, and treatment modality of the included patients

Variables	Groups				<i>p</i> value
	PBM		SHAM		
	<i>n</i> (17)	%	<i>n</i> (19)	%	
Gender					1.0000
Male	13	76.47	15	78.95	
Female	04	23.53	04	21.05	
Age group (years)					0.4768
20–39	03	17.64	02	10.52	
40–59	07	41.18	12	63.16	
60–79	07	41.18	05	26.32	
Marital status					0.9153
Single	04	23.53	04	21.05	
Married	11	64.71	14	73.68	
Separated	01	05.88	0	0	
Widowed	01	05.88	01	05.27	
Skin color					0.8952
White	02	11.76	02	10.52	
Brown	09	52.94	08	42.10	
Black	05	29.42	08	42.10	
Indigenous	01	05.88	01	05.27	
Education					0.7342
Illiterate	0	0	02	10.52	
Incomplete basic education	07	41.18	08	42.10	
Complete basic education	03	17.64	03	15.79	
Complete high school	07	41.18	06	31.58	
Family income (minimum wage)					0.7523
Up to 01	05	29.42	09	47.37	
From 02 to 05	06	35.30	06	31.58	
None	03	17.64	02	10.52	
Social security	03	17.64	02	10.52	
Alcohol consumption					0.3911
Yes	13	76.47	17	89.47	
No	04	23.53	02	10.53	
Smoking					0.7060
Yes	12	70.58	15	78.95	
No	05	29.42	04	21.05	
Alcohol consumption + smoking					0.5593
Yes	11	64.70	14	73.68	
No	06	35.30	05	26.32	
Tumor location					0.7703
Oral cavity	04	23.53	06	31.58	
Oropharynx	07	41.18	06	31.58	
Larynx	05	29.42	04	21.05	
Other (parotid, rhinopharynx)	01	05.88	03	15.79	
Tumor staging					0.1744
I	03	17.64	03	15.79	
II	01	05.88	03	15.79	
III	04	23.53	01	05.27	

Table 1 (continued)

Variables	Groups				<i>p</i> value
	PBM		SHAM		
	<i>n</i> (17)	%	<i>n</i> (19)	%	
IV	05	29.42	11	57.89	
Undefined	04	23.53	01	05.27	

and a BMI reduction of 5.71 kg (± 3.45) ($p < 0.001^*$) and 2.11 kg/m² (± 1.23) ($p < 0.001^*$), respectively (Fig. 1).

To calculate the difference between the 1st and 24th radiotherapy sessions to compare the two groups, the weight and the delta BMI were used. It was observed that the PBM group had a significantly lower weight loss (2.72 kg $\pm$ 2.2) compared to the SHAM group (5.71 kg $\pm$ 3.45) ($p = 0.004^*$). When the BMI was evaluated, the PBM group showed a minor reduction (1.05 kg/m² $\pm$ 0.83) when compared to the SHAM group (2.11 kg/m² $\pm$ 1.23) ($p = 0.005^*$) (Table 2).

When evaluating OM, it was observed that a total of 82.4% of the individuals in the PBM group had the condition, while in the SHAM group, 68.4% were affected by this side effect ($p = 0.451$). In the analysis of OM interference on weight and BMI, patients affected by this side effect showed a more significant weight loss (4.85 kg $\pm$ 3.1) and BMI reduction (1.14 kg/m² $\pm$ 1.89) when compared to those without OM, weight loss of 2.65 kg ($\pm$ 3.3) and a BMI reduction of 1.03 kg/m² ($\pm$ 1.59). In patients with mucositis, the weight loss, represented by the delta weight, was 1.9 kg (0.7–7.9) in the PBM group and 5.8 kg (5.1–7.9) in the SHAM group ($p = 0.87$). As for the BMI (delta BMI), a 0.69 kg/m² (0.47–2.9) BMI reduction was seen in the

Table 2 The influence of photobiomodulation on body weight (kg) and BMI (kg/m²) between the 1st and 24th radiotherapy sessions

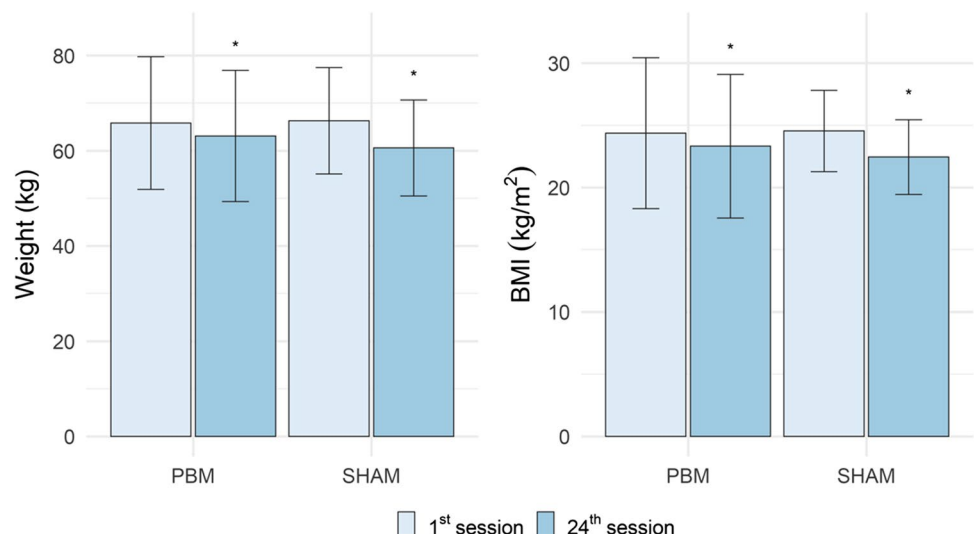
Groups	Delta weight	<i>p</i> value	Delta BMI	<i>p</i> value
PBM	2.72 kg ($\pm$ 2.2)	< 0.001*	1.05 kg/m ² ($\pm$ 0.83)	0.005*
SHAM	5.71 kg ($\pm$ 3.45)		2.11 kg/m ² ($\pm$ 1.23)	

PBM group, and 2.18 kg/m² (1.9–2.94) in the SHAM group ($p = 0.87$).

Patients allocated into the PBM group presented with less severe cases of OM, mainly grades 1 and 2, in comparison with the SHAM group, that showed a higher frequency of grades 2 (oral erythema, ulcers, with solid diet tolerated) and 3 (oral erythema, ulcers, liquid diet only) ($p = 0.004^*$).

We also evaluated the SIWL throughout the 24 radiotherapy sessions, corresponding to approximately 1 month. Participants of the PBM group showed a weight loss percentage of 4.7%, classified as acceptable. In the SHAM group, however, the percentage of weight loss was 8.5%, which determines a severe weight loss. Regarding the BMI classification, changes were observed throughout the treatment.

For the initial BMI evaluation, the PBM group had 41.7% of the individuals within the normal weight/adequate weight category and 23.5% underweight. In contrast, in the SHAM group, 47.4% of participants were in the overweight category and 36.8% in the normal weight/adequate weight category, with no statistically significant difference between groups ($p = 0.216$). As for the final BMI evaluation, in the PBM group, 47.1% of the patients were classified in the eutrophic/adequate weight category and 29.4% in the low weight category, whereas in the SHAM group, 73.7% of patients were in the normal/adequate weight category, and

Fig. 1 Body weight loss in kilograms (kg) and BMI (kg/m²) reduction according to PBM use or not, between the 1st and 24th sessions, represented by the mean and standard deviation

*Statistically differs from 1st session ($p < 0.001$)

15.8% were classified in the low weight category. The difference, however, was not statistically significant between groups ($p=0.259$). For the overall BMI analysis encompassing individuals of both groups, it was observed that prior to the cancer treatment, most individuals were eutrophic/adequate weight (38.9%) and overweight (33.3%). When the same BMI analysis took place (final analysis), the eutrophic/adequate weight category represented 61.1% of the sample and the low weight category, 22.2% (Table 3).

Regarding tumor size, patients with tumors in advanced stages (stage 03–04) had a higher SIWL compared to individuals with tumors in early stages and tumors with undetermined stages. In the PBM group, ten patients had expected SIWL, and seven patients had severe SIWL. In the SHAM group, six individuals developed an expected weight loss and 13 participants developed severe weight loss. There was no statistically significant difference between the groups ($p=1.000$) (Table 4).

As for the interference of OM severity on SIWL, it was observed that individuals who presented a moderate OM had a higher SIWL, a severe condition when compared to those with mild OM, which was considered expected weight loss. In the PBM group, seven patients developed severe SIWL, and 10 developed expected SIWL. As for the SHAM group, a total of 13 individuals presented severe weight loss, while only six participants experienced expected SIWL. There

was a statistically significant difference in both groups ($p=0.001^*$ and $p=0.018^*$) (Table 4).

Discussion

In this study, we evaluated the therapeutic effects of PBM for chemoradiotherapy-induced OM and its clinical implications on nutritional status in patients diagnosed with HNC. This study is innovative as it aims to demonstrate the beneficial role of PBM in the negative nutritional impacts caused by OM.

The socioeconomic aspects that may interfere with weight loss and BMI reduction are tumor location, tumor staging, and the therapeutic regimen. Based on the significant homogeneity between the PBM and SHAM groups regarding their socioeconomic aspects, it was inferred that it did not influence BMI reduction and weight loss outcomes.

Weight loss is often unavoidable in the cancer population [19]. It affects 80% of HNC patients in a rapid manner. Thus, one of the primary outcomes of this condition is malnutrition [20]. Advanced staging and combined therapeutic regimens are more likely to have more significant negative nutritional impacts, impairing, or incapacitating oral food intake. Additionally, advanced staging tumors can cause obstructions in the oral feeding route because of their

Table 3 Photobiomodulation influence on BMI* classification at 1st and 24th radiotherapy sessions

Initial BMI (categories)	PBM		SHAM		Total		<i>p</i> value
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	
Eutrophic/adequate weight	07	41.2	07	36.8	14	38.9	0.216
Overweight	03	17.6	09	47.4	12	33.3	
Obesity	03	17.6	01	5.3	04	11.1	
Underweight	04	23.5	02	10.5	06	16.7	
Final BMI							
Eutrophic/adequate weight	08	47.1	14	73.7	22	61.1	0.259
Overweight	02	11.8	02	10.5	04	11.1	
Obesity	02	11.8	-	-	02	5.6	
Underweight	05	29.4	03	15.8	08	22.2	

*According to the WHO (2006) and PAHO (2002) criteria

Table 4 Influence of tumor size and oral mucositis severity on the rate of involuntary weight loss (SIWL) in patients submitted or not to PBM, expressed as absolute frequency

Tumor size	PBM		<i>p</i> value	SHAM		<i>p</i> value
	Expected SIWL	Severe SIWL		Expected SIWL	Severe SIWL	
Not determined	03	01	0.054	00	01	1.000
Stage 01–02	03	01		02	04	
Stage 03–04	04	05		04	08	
Severity of mucositis						
Mild (0–1)	09	00	0.001*	04	02	0.018*
Moderate (2–3)	01	07		02	11	

extension and have higher energetic and metabolic demands for their growth [21]. By acting locally and systemically, the combination therapy can induce or exacerbate potentially harmful side effects, including OM, xerostomia, poor oral health, malabsorption, constipation, diarrhea, emesis, nausea, dysgeusia, dysphagia, and pain [22]. These are considered predisposing factors for weight loss and lead to worsening of the nutritional status and prognosis, and a decrease in quality of life [23, 24].

For the individuals affected by OM, it was observed that there was a deficit in body weight and BMI. Through a similar methodology, Goobo et al. (2014) [13] demonstrated comparable results regarding the use of PBM in the prevention and treatment of OM and its effects on body weight and BMI in HNC patients. However, the laser equipment used by the authors had higher power and a wavelength of 970 nm. The authors used the laser unfocused to obtain biomodulatory effects. Still, they did not indicate whether a device was used for this purpose or the distance between the laser tip and the oral mucosa. MASCC/ISOO panel recommends using a low-power laser to prevent and treat OM due to its biological properties: anti-inflammatory, analgesic, and wound healing, without any significant adverse effects [11]. In our study, PBM could not prevent OM, but a reduction in its severity was observed. The greater the severity of OM, the more significant the interference with the individual's ability to feed orally and the more significant impact on the nutritional status.

According to Sonis (2007) [25], OM is an inflammatory reaction characterized by cytokines with a pro-inflammatory profile. Thus, the greater the pro-inflammatory gradient concentration, the more significant painful symptomatology and, consequently, non-beneficial impacts on body weight and BMI. It is hypothesized that PBM reduces the severity of OM by decreasing the levels of inflammatory cytokines, therefore preventing the progression to more advanced stages. The findings of severe SIWL for the patients in the SHAM group indicate that they were more vulnerable to the effects of OM than the individuals in the PBM group. The evaluation of SIWL is an essential parameter for understanding weight variations over time since it can measure the severity of weight loss, identifying the risk and the various stages of malnutrition [8, 16].

The recommended PBM protocol to prevent OM requires using a low-power laser with a wavelength of 660 nm, energy density of 6.2 J, in 72 application points, 10 s per point, for 03 days a week on alternate days [11]. Studies in the literature [26–30] reported a variety of PBM protocols, with energy density ranging from 1 to 6 J, using 09 to 58 application points, being applied daily or on alternate days, and using different conductive media. Previous data from our group [31], that used the following PBM protocol; low-power AsGaAl laser, 660 nm, 86.7 mW, the

energy density of 2 J, 28 application points in the oral cavity for 3 s each point was also unable to prevent OM; however, as observed in this study, there was a reduction in the severity of OM. These laser specifics and different protocols render it challenging to compare the studies but do not invalidate the relevance of their findings.

The anthropometric measurements, mainly through assertive and prompt methods such as measuring body weight, checking the variation of body weight, BMI calculation, and its classification are extremely important for the early diagnosis of cancer patients' most frequent nutritional disorders. This rapid and easy identification can help in counseling and identifying possible nutritional interventions preventing nutritional damage, thus providing a better prognosis for the patients [13, 22].

Previous studies [23, 24] have demonstrated a strong correlation between pretreatment BMI and weight loss; however, the findings are conflicting as they show divergences concerning the BMI classification. The first study states that individuals classified as overweight before antineoplastic therapy are more likely to maintain or have a minor reduction in body weight during treatment. It reduces possible treatment interruptions due to their immune function not being compromised compared to individuals with low weight and obesity. Thus, overweight individuals tend to present better outcomes for quality of life and future nutritional status. In contrast, for Zhao et al. (2015) [33], pretreatment overweight or obesity was a predisposing factor for weight loss compared to eutrophic individuals. The finding was explained by professionals' tendency to prescribe nutritional supplements for underweight and eutrophic patients, aiming at bodyweight recovery or maintenance. This practice, however, is not performed with overweight and obese patients, making their weight loss more evident.

Weight loss is a sign of advancing malnutrition in patients diagnosed with HNC and needs to be detected early on the treatment. Being aware of the SIWL before and during treatment is extremely important, as high scores are strongly associated with negative impacts on treatment and quality of life and increased morbidity and mortality in cancer patients [32]. The study by Farhangfar et al. (2014) [33] correlated the SIWL with nutritional impact symptoms, such as dysphagia, mucositis, masticatory problems, anorexia, loss of appetite, depression, among others symptoms, in 635 patients with HNC mostly diagnosed with advanced-stage tumors (65.5%). However, the SIWL was assessed according to the history of weight change 6 months before cancer treatment. Thus, it was observed that the total score of symptoms of nutritional impact ($p < 0.0001$) as well as tumor stage ($p = 0.006$) were significant predictors for SIWL. There is a lack of studies correlating SIWL with tumor size and severity of

OM during oncologic therapy. Our study was pioneering on demonstrating a direct correlation between weight loss in patients with higher levels of mucositis.

Our study had limitations due to the sample size, which may have interfered with the absence of statistical significance when evaluating the influence of PBM on the initial and final body weight and BMI differences according to the presence or absence of OM, as well as in the classification of BMI.

Nevertheless, the numerical difference between the group of patients with OM who underwent PBM and the control group was significant, both for differences in weight loss and BMI. It was not possible to obtain data on enteral nutrition, which is also a limitation of the present study. Cancer treatment has some particularities, such as interruption due to worsening side effects, indisposition, and death, which end up substantially reducing the sample size.

Conclusion

PBM for OM reduced weight loss and prevented BMI reduction in patients undergoing chemoradiotherapy for HNC.

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Author contribution This is an original study, and all authors contributed to the study's conception, design, and approved the final version of the manuscript. Amanda de Sousa Melo and Camila Lima Andrade participated in manuscript writing and the experiment's execution. Juliana Borges de Lima Dantas, Alena Ribeiro Alves Peixoto Medrado, Gabriela Botelho Martins, and Hayana Ramos Lima collaborated in the experiment's execution, manuscript review, English proofreading, and critical review. Manoela Carrera supervised all stages of the work and participated in the manuscript writing and review.

Data Availability <https://data.mendeley.com/datasets/4kd7s49wk4/1>, reference [<https://doi.org/10.176332/4kd7s49wk4.1>].

Code availability Not applicable.

Declarations

Ethics approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and conducted per the Declaration of Helsinki. CAAE: 68263317.8.0000.0047.

Consent to participate Informed consent was obtained from all participants included in the study.

Consent for publication Not applicable.

Conflict of interest The authors declare no competing interests.

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